A Sri Ganesh

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.33
Narayana Junior College	Class XII (Telangana State Board of Intermediate Education)	2019	91.30%
Narayana Concept School	Class X (Board of Secondary Education Telangana State)	2017	9.00%

SCHOLASTIC ACHIEVEMENTS

- **Patent** application filed with Intellectual Property India for a novel Dynamic Vision Sensor (DVS) architecture.
- Finalist in Toycathon 2021, organized by the Ministry of Education's Innovation Cell, New Delhi.

PUBLICATIONS

- Proposed and designed "A Novel Low-Power, High-Resolution, and High-Speed Dynamic Vision Sensor for Event Detection," accepted for presentation at the **IEEE Asia Pacific Conference on Circuits and Systems (APCCAS) 2026**.
- Designed and published "Substantial Capabilities of 3-Axis Robotic Arm in Industry Automation Applications" in IEEE CISCT 2023 (DOI: 10.1109/CISCT57197.2023.10351387).

SKILLS

EDA Tools: Cadence Virtuoso, Spectre, Calibre

Circuit Design & Layout: Analog IC Design, Custom Layout Design, Post-Layout Analysis & Debugging.

Validation: Hands-on Post-Silicon Validation, Test Automation

Programming: Verilog HDL, Python, C

WORK EXPERIENCE

Research Assistant, IC Design Lab, Department of Electrical Engineering, IIT Hyderabad

Advisor: Dr. Kapil Jainwal

- Designed CMOS Image Sensors, including 2D Image Sensors, Dynamic Vision Sensors (DVS), and 3D Image Sensors.
- Executed **schematic design**, **custom layout**, and **post-layout simulation/analysis** to debug the designed sensors.
- Proposed **novel sensor architectures**, leading to patent applications and an IEEE conference publication.

Product Engineer, DDR PHY IP Group, Cadence Design Systems, Bangalore(Internship)

- Performed **post-silicon validation** of **DDR PHY** IP, including calibration, electrical, and signal-integrity characterization.
- Conducted **hands-on** high-speed interface characterization across **LPDDR5x** and **GDDR6** using automated test

setups.
- Developed **automated data collection** and analysis workflows using Python and C.

PROJECTS

Master Thesis: Design of CMOS Image Sensors and Event-Based Vision Sensors

Guide: Dr. Kapil Jainwal

High Dynamic Range 2D Image Sensor | *Jan 2026 - Jul 2026*

- Designed a **full-custom** 4T pixel with 5.7 µm pitch, CDS bias circuits, and **10-bit DAC** for HDR image acquisition.
- Developed a 272x256 pixel array and **taped out** the sensor in 180 nm X-FAB CMOS technology for DRDO (IRDE).

Novel Dynamic Vision Sensor for High-Speed Event Detection | *Sep 2025 - Dec 2025*

- Proposed a compact, scalable 5T-**DVS architecture** with column-level event detection, achieving 7 µm pitch, 40% fill factor, and reduced pixel complexity.
- **Taped out** a 256x256 DVS array in SCL 180 nm CMOS with 81.92 µs event latency and 3.125 M events/s.

SPAD-Based 3D Depth Sensor for Long-Range Detection | Mar *2025 - Aug 2025*

- Designed a SPAD-based depth sensor for long-range detection up to 750 m using a hybrid **TDC-TAC** (Time-to-Digital/Time-to-Amplitude Converter) architecture.
- Developed readout circuitry for long-range depth extraction and **taped out** the test circuit in SCL 180 nm CMOS technology.

2-Stage Operational Transconductance Amplifier (OTA)

- Designed and optimised a two-stage Miller-compensated OTA with single-ended output in TSMC 180 nm CMOS technology.
- Achieved 63.54° phase margin, 5.9 V/µs slew rate, and 0.2-1.08 V ICMR at 1.8 V supply operation.

RELEVANT COURSES

- **Analog Domain** : Analog IC Design, Analog IC Lab, Mixed Signal Circuit Design.
- **Digital Domain**: Introduction to VLSI Design, Digital IC Design, Digital IC Design Lab.
- **Memory & Device Domain** : Memory Circuits and System Design, Semiconductor Device Modelling, VLSI Technology.

POSITIONS OF RESPONSIBILITY

- **Placement Coordinator**, Department of Electrical Engineering, IIT Hyderabad - Coordinated with recruiters and students, facilitating placement activities and supporting the recruitment process.
- **Teaching Assistant**, *Analog Electronic Circuits* and *CMOS VLSI Design*, IIT Hyderabad - Mentored UG and PG students and managed course logistics under the instruction of **Dr. Kapil Jainwal**.

EXTRACURRICULAR

- Chess, Volleyball, Cooking and strength training.